KYUNGMIN LEE

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PROFESSIONAL APPOINTMENT

Assistant Professor, Adelphi University, Garden City, NY, USA — Aug 2026 – Present
Department of Environmental Studies and Sciences

Sustainability Director, Adelphi University, Garden City, NY, USA — Aug 2026 – Present
Office of Sustainability

EDUCATION

Ph.D., Energy and Environmental Policy, University of Delaware, Newark, DE, USA — 2025

Master of City Planning, Environmental Studies, Seoul National University, Seoul, South Korea — 2018

B.A., Economics, Sungshin Women's University, Seoul, South Korea — 2014

PUBLICATIONS

† corresponding author, ∗ equal contribution

Peer-Reviewed Journal Articles

5. Wijaya, B. N., Park, Y., Jeung, J. H., & **Lee, K**†. (2026). Exploring the relationship between air quality and happiness in South Korea using artificial neural networks. *Environmental Impact Assessment Review*, 117, 108135. https://doi.org/10.1016/j.eiar.2025.108135

4. **Lee, K.**, Yun, S.J. †, & Myeong, J.Y. (2026) Proposing a Value-Sharing Mechanism for Wind Power: Impact of Benefit-sharing and Community Engagement on Acceptance and Environmental Awareness. *Energy Research & Social Science.* https://doi.org/10.1016/j.erss.2026.104537

3. **Lee, K.**, Yun, S.J. †, & Park, S.A. (2025) Varied Drivers for Reducing CO_2 Emissions in Countries with Legislated Carbon Neutrality Targets: A Kaya Identity Clustering Analysis. *Environment, Development and Sustainability.* Doi: https://doi.org/10.1007/s10668-025-06223-2
    - Awarded the Best Student Paper Award from the NECoPA 2023 Fall Conference.
    - Awarded the 2022 Excellence Research Award from the Korea Safety Health and Environment Foundation.

2. Arjunan, P., Dobler, G., **Lee, K.**, Miller, C., Biljecki, F., & Poolla, K. † (2021, November). Operational characteristics of residential air conditioners with temporally granular remote thermographic imaging. In *Proceedings of the 8th ACM international conference on systems for energy-efficient buildings, cities, and transportation* (pp.184-187). Doi: https://doi.org/10.1145/3486611.3486659

1. **Lee, K.**, & Yun, S.J.† (2018). Positive Effects and Problematic Conditions of Adopting the Benefit Sharing Mechanism in Renewable Energy Projects: Focusing on the Case of Onshore Wind Farms in Jeju Island. *Space & Environment*: Vol.65. (pp.227~270). [In Korean] Doi: https://doi.org/10.19097/kaser.2018.28.3.227
    - Featured in the *Monthly Electrical Technology & Information* 2019 January [CrossRef]

Under Reviewed

1. **Lee, K.**, Sharma, M.S., Karp, A., & Dobler, G. † The Dependence of End Use Behavior of Cooling Systems on Environmental Conditions.
   - Reviewed and Resubmitted to *Energy and Buildings*

Work in Progress

2. **Lee, K.**, & Dobler, G. † A Machine Learning Approach to Mapping Urban Heat Resilience with Satellite Image Analysis in Wilmington, Delaware.
   - *Preparing for submission*
1. **Lee, K.**, Powers, T., & Dobler, G.† Privacy-preserving Spatio-temporal Data Aggregation for Energy-End Use in City.
   - *Preparing for submission*

Policy Reports

4. Cho, H., Lim, Y., Kim, S., Jeong, H., & **Lee, K.** (2019) Technical Assistance for the Development of an Urban Adaptation Plan for Kurunegala Project, D4.1 Report on the Results of Climate Change Risk and Vulnerability Assessment. CTCN Requests No. 2018000009 Vol 3. UN Climate Technology Centre and Network, & Korea Environment Institute.
3. Cho, H., Rubio, C., Jeong, H., **Lee, K.**, & Odawa, S. (2019) Technical Assistance for the Development of an Urban Adaptation Plan for Kurunegala Project, D4.2 Climate Change Risk Assessment Guidelines. CTCN Requests No. 2018000009 Vol 4. UN Climate Technology Centre and Network, & Korea Environment Institute.
2. Kim, G., & **Lee, K.** (2016). A Comparative Study of Humanitarian Programs for Climate Refugees. *Enerzine Focus*: Vol.69 Energy Climate Policy Institute; Republic of Korea.
1. Kim, T., Maeng, J., Park, J., & **Lee, K.** (2014). A Study on Environmental Assessment Guideline for Site Selection of Offshore Wind Farm Project. Korea Environment Institute.

(SELECTED) CONFERENCE PRESENTATIONS

Conference Panel Paper/Poster Presentations

2026
- A Machine Learning Approach to Mapping Heat Resilience in Wilmington, Delaware. *American Society for Public Administration (**ASPA**).* Los Angeles, USA. Panel Presentation. March 2026.
- Value-Sharing Mechanisms for Local Governments, Wind Power Companies, and Civic Engagement. *American Society for Public Administration (**ASPA**) SPOD Session.* Los Angeles, USA. Panel Presentation. March 2026.

2025
- Revisiting Easterlin Paradox: Main Drivers of Sustainability for Subjective Happiness in South Korea 2020-2023. *Public Management Research Conference (**PMRC**).* Seoul, South Korea. Poster. June 2025. (Accepted)
- Did Finance and Engagement Increase Acceptance and Environmental Awareness? *American Society for Public Administration (**ASPA**).* Washington DC, USA. Panel Presentation. March 2025. (Accepted)

2024
- A Novel Approach to Assessing the Efficacy of Policy-driven Urban Energy Demand Reduction via Proximal Infrared Remote Sensing. *Conference on Neural Information Processing Systems* (***NeurIPS**) 2024 Workshop on Tackling Climate Change with Machine Learning.* Vancouver, Canada. Poster. December 2024.
- Insight to Action: Optimizing Carbon Emissions and Cost Reductions for Household Energy Demand Using Machine Learning. *Conference on Neural Information Processing Systems* (***NeurIPS**) 2024 Workshop on Tackling Climate Change with Machine Learning.* Vancouver, Canada. Poster. December 2024.
- Behavioral Metrics and the Limits of Aggregation for Anonymization and Privacy

Preservation Policy. *Association for Public Policy Analysis & Management (**APPAM**).* National Harbor, Maryland, USA. Panel Paper. November 2024.

- Monitoring HVAC Use Behavior in Buildings and Energy Policy Using Proximal Infrared Images. *Association for Public Policy Analysis & Management (**APPAM**).* National Harbor, Maryland, USA. Poster. November 2024.
- Did Financial Benefit-sharing and Community Engagement Increase Residential Acceptance of Windmills and Environmental Protection Awareness? *Association for Research on Nonprofit Organizations and Voluntary Action (**ARNOVA**)*. Washington DC, USA. Panel Paper. November 2024.
- Monitoring HVAC Use and Behavioral Metrics for Energy and Privacy Policy Using Proximal Infrared Images. *Chungnam National University Conference of Advanced Technology (**CCAT**)*. Virtual Presentation. Panel Paper. November 2024.
- Proposing a Value-Sharing Mechanism for Wind Power: Impact of Benefit-sharing and Community Engagement on Acceptance and Environmental Awareness. *Northeast Conference on Public Administration (**NECoPA**).* New York City, USA. Panel Paper. November 2024.
- Analyzing Benefit-Sharing Mechanisms for Wind Farms in Jeju Island: A Framework and Survey. *Association for Research on Nonprofit Organizations and Voluntary Action (**ARNOVA**)-Asia*. Seoul, South Korea. Poster. June 2024.
- Privacy-Preserving Spatio-Temporal Data Aggregation for Energy End Use in City. *Urban Association Affairs (**UAA**).* New York City, USA. Panel Paper. April 2024.

2023

- A Novel Approach to Assessing the Efficacy of Policy-driven Urban Energy Demand Reduction via Proximal Infrared Remote Sensing. *Association for Public Policy Analysis & Management (**APPAM**).* Atlanta, Georgia, USA. Panel Paper. November 2023.
- Drivers of CO2 Emissions Trends in Countries That Have Legislated Carbon Neutrality Targets. *Northeast Conference on Public Administration (**NECoPA**).* Virtual Presentation. Panel Paper. November 2023.

2022

- An Exploratory Study on Carbon Neutrality Monitoring Models and Scenarios for the Establishment of a Sustainability Assessment System: A Comparison of Korea and Countries Enacted Legally Binding Agreements. *Korea Safety, Health, and Environment Foundation*, Virtual Presentation. Panel Paper. November 2022.

2018

- An Influence of the Benefit Sharing Mechanism to the Local Acceptance and Environmental Awareness: Focusing on the Case of Onshore Wind Farms in Jeju Island. *Bio-resources, Energy, Environment, and Materials Technology International Conference*. Hongcheon, South Korea. Poster. June 2018.
- An Exploratory Study on Positive Effects of and Problematic Conditions of Adopting the Benefit Sharing Mechanism: Focusing on the Case of Onshore Wind Farms in Jeju Island. *Korea Environmental Policy and Administration Society*. Seoul, South Korea. Panel Paper. February 2018.

2016

- A Study on Impacts from Price Subsidy and Tax Policy of EVs in South Korea. *Korea Environmental Economics Association International Conference*. Seoul, South Korea. Panel Paper. June 2016.

Invited Lectures / Guest Speakers

2025

- Applications of Deep Learning and Remote Sensing in Energy and Environmental Policy. *School of Digital Humanities and Computational Social Science. Korea Advanced Institute of Science and Technology (KAIST).* November 2025.

2024

- Novel Approach to Assessing the Efficacy of Policy-driven Urban Energy Demand Reduction via Proximal Infrared Remote Sensing. *Principles of Urban Data Science (SPPA720). Biden School of Public Policy and Administration. University of Delaware*. September 2024.

2023
- Computational Social Science for Energy and Environmental Policy. *Department of Public Administration. Chungnam National University*. October 2023.
- Data-Driven Policy Making For Disaster Management. *Department of Public Administration. Chungnam National University*. April 2023.

2021
- Overview of Data Science and Machine Learning Tools in the Energy and Environmental Policy Field. *Department of Public Administration. Chungnam National University*. November 2021.

2020
- A Case Study of Machine Learning Applications in Energy and Environmental Policy. *Department of Public Administration, Chungnam National University*. October 2020.

HONORS AND AWARDS

Scholarships & Fellowships

2023-24 University of Delaware, Doctoral Fellowship for Excellence (~$30,000)
2022 University of Delaware, Summer Doctoral Fellowship ($4,500)
2019-23 University of Delaware, Tuition & Assistantships
2015-16 Seoul National University, Merit-based & Work-Study Scholarships
2011-13 Sungshin Women's University, Merit-based Scholarships

Grants & Awards

2025 Peter A. Larson Prize ($1,000)
2024 Third Place Student Paper Award, Korean Association for Policy Studies (**KAPS**)-Association for Public Policy Analysis & Management (**APPAM**) ($50)
UD Travel Grant Awards ($3,000)
2023 Best Student Paper Award, Northeast Conference on Public Administration (**NECoPA**) ($140)
2022 Research Excellence Award, Korea Safety Health and Environment Foundation (**KSHEF**) Environmental Research Forum ($250)
2015 Research Grant Competition Award, Korea Energy Climate Policy Institute (**KECPI**) ($1,550)

Government & International Recognitions

2024 Climate Change AI In-Person Summer School ($1,000)
2022 Aspen Institute, Future Leaders Climate Summit Cohort (Travel Award)
2016 Ministry of Environment of Korea & Korea Environment Corporation, Global Environmental Leaders Program, UN Internship Support (~$6,000)
2016 Korea Meteorological Administration, International Meteorological Expert Program, UN Internship Support (~$3,600)
2014 Korea National Assembly Climate Change Forum, Climate Change Academy, Excellence Award (Travel Award, UNFCCC COP20 Observer)

TEACHING EXPERIENCE

Adelphi University
Undergraduate-Level Courses, Instructor
Natural Science and Environmental Problems (EAS110 & ENV102) Fall 2026

University of Delaware
Graduate-Level Course, Adjunct Faculty Instructor
Machine Learning for Public Policy (SPPA722) Spring 2026
Undergraduate Level Courses, *Teaching Assistant/Guest Speaker* Aug 2019 – Dec 2022
Crafting Public Policy (UAPP225)
Non-Governmental Actors: Interest Groups Spring 2022
Environmental and Energy Policy Spring 2021
Policy Implementation and Regulatory Rule-Making Spring 2020

Introduction to Energy and Environmental Policy (ENEP250)
Global Environmental Law and Policy/Climate Policy/Seminar — Fall 2022
Study Team Workshop — Spring 2022
Environmental Law and Policy — Fall 2020
Political Economy of the Environment (ENEP410)
Global Capitalism and the Environment — Fall 2021
Environmental Sustainability: Political and Economic Analysis (ENEP410)
Global Financing: Climate Change and the Least Developed Countries — Fall 2019

PROFESSIONAL EXPERIENCE

Research Fellow, **University of Connecticut**, USA — Oct 2025 – Aug 2026
Research/Teaching Assistant, **University of Delaware**, USA — Aug 2019 – Aug 2025
Research Assistant, **Korea Environment Institute**, South Korea — Apr 2019 – May 2019
International Affairs Specialist, **Korea Marine Environment Management Corporation**, South Korea — Jun 2018 – Sep 2018
Green Growth Knowledge Platform Intern, **United Nations Environment Programme (UNEP),** Switzerland — Mar 2017 – Aug 2017
Environmental Affairs Intern, **United Nations Economic and Social Commission for Asia and the Pacific-East and North-East Asia (UNESCAP-ENEA),** South Korea — Sep 2016 – Feb 2017
Research Assistant, **Seoul National University**, South Korea — Sep 2015 – Aug 2016
Research Intern, **Korea Environment Institute**, South Korea — Aug 2014 – Dec 2014

SERVICE ACTIVITIES

Journal Review
2026 Energy Policy, Natural Hazards
2025 Quality & Quantity
2023 Policy Studies Journal

Conference Service
ASPA Section on Professional and Organizational Development, Board Member at Large & Treasurer, 2025-2027
ARNOVA-Asia, Session Moderator, June 2024
APPAM Data Science Community, Member, 2023-2024

Department and University Service
UD Climate Change Hub, Climate Fellow, 2023-2025
UD Data Science Institute, Graduate Fellow, 2021-2025
UD Computational Social Science Association, Interim President & Vice President, 2022-2023
UD Biden School Student Association, Secretary, 2021-2022
UD Korean Graduate Student Association, Officer, 2021-2022
UD Energy and Environmental Policy Student Association, Alumni Chair, 2020-2021

Community Service
BigWave - Korean Youth Climate Change Network, Co-founder & Steering Committee, 2016-2019

TRAINING

*Summer Institute in Computational Social Science (SICSS)-Korea 2024***,** KAIST-KDI School of Public Policy and Management
Climate Change AI In-Person Summer School 2024, Mila-Québec AI Institute

LANGUAGES

English (Fluent), Korean (Native)

SKILLS

***Programming & Tools*:** Python (Main), R, STATA, SPSS, ArcGIS, LaTeX
***Quantitative Methods*:** Artificial Intelligence/Computer Vision (Main)**,** Geospatial Analysis, Statistics
***Qualitative Methods*:** Semi-structured Survey, In-depth Interviews